Artificial Intelligence & Generative AI

Question & Answer Reference Dataset for RAG Systems

28 Curated Q&A; Pairs | 5 Topic Sections

1. Fundamentals of Artificial Intelligence

Q1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is the simulation of human intelligence processes by computer systems. It involves the development of algorithms and models that enable machines to perform tasks that typically require human intelligence, such as reasoning, learning, problem-solving, perception, and language understanding.

Tags: AI, Definition, Fundamentals

Q2. What is the difference between Narrow AI and General AI?

Narrow AI (Weak AI) is designed to perform a specific task, such as facial recognition or language translation. It excels in its domain but cannot generalize beyond it. General AI (Strong AI) refers to a hypothetical system that can perform any intellectual task that a human can do — it possesses flexible, cross-domain reasoning capabilities. As of today, all deployed AI systems are Narrow AI.

Tags: Narrow AI, AGI, General Intelligence

Q3. What are the main branches of AI?

The main branches of AI include: (1) Machine Learning — systems that learn from data; (2) Natural Language Processing (NLP) — understanding and generating human language; (3) Computer Vision — interpreting visual information; (4) Robotics — physical AI agents; (5) Expert Systems — rule-based decision making; and (6) Planning and Optimization — AI for scheduling and problem solving.

Tags: ML, NLP, Computer Vision, Robotics

Q4. What is Machine Learning (ML)?

Machine Learning is a subset of AI where systems automatically learn and improve from experience without being explicitly programmed. Instead of writing rules manually, ML algorithms identify patterns in data and build models that can make predictions or decisions on new, unseen data.

Tags: Machine Learning, Supervised, Unsupervised

Q5. What are the types of Machine Learning?

There are three main types: (1) Supervised Learning — the model is trained on labeled data (e.g., classification, regression); (2) Unsupervised Learning — the model discovers hidden patterns in unlabeled data (e.g., clustering, dimensionality reduction); (3) Reinforcement Learning — an agent learns by interacting with an environment and receiving rewards or penalties.

Tags: Supervised, Unsupervised, Reinforcement Learning

2. Deep Learning & Neural Networks

Q6. What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses multi-layered artificial neural networks (deep neural networks) to model complex patterns in data. The term 'deep' refers to the many layers in the network. Deep Learning powers modern applications like image recognition, speech synthesis, and large language models.

Tags: Deep Learning, Neural Networks, Layers

Q7. What is a neural network and how does it work?

A neural network is a computational model inspired by the human brain. It consists of nodes (neurons) organized in layers: an input layer, one or more hidden layers, and an output layer. Each neuron receives inputs, applies a weighted sum, passes it through an activation function, and sends the output forward. During training, weights are adjusted using backpropagation and gradient descent to minimize prediction error.

Tags: Neural Network, Backpropagation, Weights

Q8. What is a Transformer architecture?

The Transformer is a deep learning architecture introduced in the 2017 paper 'Attention Is All You Need'. It relies entirely on self-attention mechanisms to process sequences in parallel, making it highly efficient for NLP tasks. Transformers are the backbone of modern large language models (LLMs) like GPT, BERT, and Claude.

Tags: Transformer, Attention, LLM, GPT, BERT

Q9. What is the attention mechanism in Transformers?

The attention mechanism allows a model to focus on relevant parts of the input when producing an output. In self-attention, each token in a sequence looks at all other tokens to compute a weighted representation. This helps the model understand relationships and context across long distances in text, which was a key limitation of older RNN-based models.

Tags: Attention, Self-Attention, Context

Q10. What are Convolutional Neural Networks (CNNs)?

CNNs are specialized neural networks designed for processing grid-like data, particularly images. They use convolutional layers to detect local features (edges, textures, shapes) by sliding learned filters over the input. Pooling layers reduce spatial dimensions, and fully connected layers make final predictions. CNNs are the standard for image classification, object detection, and segmentation tasks.

Tags: CNN, Image Recognition, Convolution, Computer Vision

3. Generative AI (GenAI)

Q11. What is Generative AI (GenAI)?

Generative AI refers to AI systems capable of creating new content — text, images, audio, video, or code — that resembles human-produced content. Unlike discriminative models that classify inputs, generative models learn the underlying distribution of training data and can generate novel outputs. Examples include GPT-4, DALL-E, Stable Diffusion, and Claude.

Tags: GenAI, Content Generation, GPT, DALL-E

Q12. What are Large Language Models (LLMs)?

Large Language Models are deep learning models trained on massive text corpora to understand and generate human language. They use the Transformer architecture and contain billions of parameters.

LLMs can perform a wide range of tasks — summarization, translation, Q&A;, code generation, and reasoning — through prompting, without task-specific fine-tuning.

Tags: LLM, Parameters, GPT-4, Claude, Gemini

Q13. What is Prompt Engineering?

Prompt Engineering is the practice of crafting effective input prompts to guide LLMs toward desired outputs. Techniques include zero-shot prompting (no examples), few-shot prompting (with examples), chain-of-thought prompting (step-by-step reasoning), and role prompting (assigning a persona). Good prompts significantly improve the quality and relevance of AI responses.

Tags: Prompt Engineering, Few-Shot, Zero-Shot, Chain-of-Thought

Q14. What is fine-tuning in the context of LLMs?

Fine-tuning is the process of further training a pre-trained LLM on a smaller, domain-specific dataset to adapt it for particular tasks or styles. It adjusts the model weights to specialize in areas like medical QA, legal reasoning, or customer support. Techniques include full fine-tuning, LoRA (Low-Rank Adaptation), and instruction fine-tuning.

Tags: Fine-Tuning, LoRA, Domain Adaptation, PEFT

Q15. What are Diffusion Models?

Diffusion models are a class of generative models that learn to create data by reversing a noise-adding process. During training, Gaussian noise is progressively added to data until it becomes pure noise. The model learns to denoise step-by-step. During inference, it starts from random noise and iteratively removes it to generate high-quality images. Stable Diffusion and DALL-E 3 use this approach.

Tags: Diffusion Models, Stable Diffusion, Image Generation

Q16. What is Retrieval-Augmented Generation (RAG)?

RAG is a technique that combines information retrieval with language model generation. When a query is received, relevant documents are retrieved from an external knowledge base (using vector similarity search) and injected into the LLM's context. This allows the model to generate accurate, grounded responses using up-to-date or domain-specific information not present in its training data.

Tags: RAG, Vector Search, Knowledge Base, Grounding

Q17. What is Hallucination in AI models?

Hallucination occurs when an AI model generates confident but factually incorrect or fabricated information. LLMs can 'hallucinate' names, dates, citations, or facts that don't exist. This happens because models generate statistically probable text, not verified facts. RAG, grounding techniques, and fine-tuning on factual data are common mitigation strategies.

Tags: Hallucination, Factuality, RAG, Grounding

Q18. What is the role of embeddings in AI?

Embeddings are dense vector representations of data (text, images, etc.) in a high-dimensional space where semantic similarity corresponds to geometric closeness. In NLP, word or sentence embeddings capture meaning — similar concepts cluster together. Embeddings are fundamental to RAG systems, semantic search, recommendation systems, and clustering tasks.

Tags: Embeddings, Vector Space, Semantic Search, RAG

4. RAG Systems & Vector Databases

Q19. What is a vector database and why is it used in RAG?

A vector database stores and indexes high-dimensional embedding vectors and supports fast approximate nearest neighbor (ANN) search. In RAG systems, documents are chunked, embedded, and stored in a vector database. When a user query arrives, its embedding is compared to stored vectors to retrieve the most semantically relevant chunks. Popular vector databases include Pinecone, Weaviate, Qdrant, Chroma, and FAISS.

Tags: Vector DB, FAISS, Pinecone, ANN Search, Chroma

Q20. What are the key components of a RAG pipeline?

A RAG pipeline consists of: (1) Document Loading — ingesting PDFs, web pages, docs; (2) Chunking — splitting documents into meaningful segments; (3) Embedding — converting chunks to vectors using an embedding model; (4) Vector Storage — indexing vectors in a vector database; (5) Retrieval — finding relevant chunks for a query; (6) Augmentation — injecting retrieved context into the LLM prompt; (7) Generation — LLM produces a final answer.

Tags: RAG Pipeline, Chunking, Retrieval, Generation

Q21. What chunking strategies are used in RAG systems?

Common chunking strategies include: (1) Fixed-size chunking — splitting by character or token count; (2) Sentence-based chunking — splitting at sentence boundaries; (3) Recursive character splitting — hierarchical splitting with overlap; (4) Semantic chunking — splitting based on topic shifts in meaning; (5) Document-structure-aware chunking — using headings, sections, and paragraphs as natural boundaries.

Tags: Chunking, Text Splitting, Overlap, Semantic Chunking

Q22. What is Cosine Similarity in the context of RAG?

Cosine similarity measures the angle between two vectors in high-dimensional space. A value of 1 means the vectors point in the same direction (high similarity), while 0 means they are orthogonal (unrelated). In RAG, cosine similarity is used to rank and retrieve the most relevant document chunks based on their embedding vectors relative to the query embedding.

Tags: Cosine Similarity, Retrieval, Ranking, Embeddings

Q23. What is Re-ranking in RAG?

Re-ranking is a post-retrieval step where initially retrieved documents are re-scored using a more powerful cross-encoder model. While vector search retrieves candidates quickly (bi-encoder), re-ranking applies a more precise relevance score by jointly encoding the query and each document. This two-stage approach improves precision. Tools like Cohere Rerank and ColBERT are commonly used.

Tags: Re-ranking, Cross-Encoder, Cohere, Precision

5. Ethics, Safety & Future of AI

Q24. What are AI Hallucinations and how can they be mitigated?

AI hallucinations are confident, false outputs. Mitigation strategies include: RAG (grounding responses in retrieved documents), fine-tuning on curated data, implementing self-consistency checks, adding citations and source attribution, using Constitutional AI (CAI) techniques, and human-in-the-loop review systems.

Tags: Hallucination, Safety, RAG, Constitutional AI

Q25. What is AI Alignment?

AI Alignment refers to the challenge of ensuring that AI systems behave in accordance with human values, intentions, and goals. A misaligned AI might pursue objectives in ways that are harmful or unintended. Alignment research includes techniques like RLHF (Reinforcement Learning from Human Feedback), Constitutional AI, interpretability research, and value learning.

Tags: Alignment, RLHF, Safety, Constitutional AI

Q26. What is RLHF (Reinforcement Learning from Human Feedback)?

RLHF is a training technique used to align LLMs with human preferences. It involves three steps: (1) supervised fine-tuning on demonstration data; (2) training a reward model on human preference comparisons between outputs; (3) using RL (typically PPO) to optimize the LLM to maximize the reward model's score. RLHF is central to models like ChatGPT and Claude.

Tags: RLHF, Reward Model, PPO, Alignment, ChatGPT

Q27. What are the ethical concerns around Generative AI?

Key ethical concerns include: (1) Misinformation and deepfakes; (2) Bias and fairness — models can perpetuate training data biases; (3) Copyright and intellectual property; (4) Privacy — models may memorize sensitive training data; (5) Job displacement; (6) Environmental impact — large models have significant carbon footprints; (7) Misuse for malicious content generation.

Tags: Ethics, Bias, Privacy, Deepfakes, Fairness

Q28. What is the future of Generative AI?

The future of GenAI includes multimodal AI systems that seamlessly handle text, image, audio, and video; more efficient models with lower compute requirements; AI agents that autonomously complete complex tasks; better alignment and safety guarantees; personalized AI assistants; integration across industries (healthcare, education, law, science); and potential progress toward Artificial General Intelligence (AGI).

Tags: Future of AI, Multimodal, AGI, AI Agents